2.10 Classwork: Systems of Equations and Exponents (no calculator) — Markscheme

Version (a)

Total 55 marks

1. [Maximum mark: 7]

Elimination is required. Substitution in place of elimination earns M0, then A1A1 for correct values. FT the second value from the first.

(a) $4x + 3y = 5 \;\; (1), \quad 6x - 5y = 17 \;\; (2)$

(i) $3 \times (1)$: $12x + 9y = 15$; $\quad 2 \times (2)$: $12x - 10y = 34$ **A1**

Both new equations needed, every term multiplied. Common error: $12x + 9y = 5$ *(left side only) earns A0.*

(ii) $(12x + 9y) - (12x - 10y) = 15 - 34$ **M1**

$19y = -19$

$y = -1$ **A1**

(iii) $4x + 3(-1) = 5$

$4x = 8$

$x = 2$ **A1**

(b) $5x + 2y = 3, \quad 3x + 7y = 25$

$3\times$ first: $15x + 6y = 9$; $\quad 5\times$ second: $15x + 35y = 125$

Subtracting: $29y = 116$ **M1**

$y = 4$ **A1**

$5x + 2(4) = 3$, so $5x = -5$ and $x = -1$ **A1**

Accept eliminating y first: $7(5x + 2y) - 2(3x + 7y)$ *gives* $29x = -29$.

2. [Maximum mark: 13]

(a) $u_1 + 3d = 17$ **A1**

$u_1 + 9d = 41$ **A1**

(b) Subtracting: $6d = 24$ **M1**

$d = 4$ **A1**

$u_1 = 17 - 3(4) = 5$ **A1**

Common error: $u_n = u_1 + nd$ *gives* $u_1 + 4d = 17$*, so* $d = 4$ *but* $u_1 = 1$*: A0 for* u_1*, and FT in (c).*

(c) $S_{20} = \frac{20}{2}(2(5) + 19(4))$ **M1**

$S_{20} = 10(86) = 860$ **A1**

FT from (b). Accept $S_{20} = 10(u_1 + u_{20}) = 10(5 + 81)$.

(d) $S_2 = p(2)^2 - q(2) = 4p - 2q$ **M1**

$4p - 2q = 2$ **AG**

(e) $25p - 5q = 35$ **A1**

Accept the simplified form $5p - q = 7$.

(f) $2p - q = 1$ and $5p - q = 7$

Subtracting: $3p = 6$ **M1**

$p = 2$ **A1**

$q = 2(2) - 1 = 3$ **A1**

(g) $u_1 = S_1 = 2(1)^2 - 3(1) = -1$ **A1**

FT from (f).

3. [Maximum mark: 9]

(a) $2A + B = 7$ **A1**

$8A + B = 19$ **A1**

(b) Subtracting: $6A = 12$ **M1**

$A = 2$ **A1**

$B = 7 - 2(2) = 3$ **A1**

(c) $y = 2\,(2^5) + 3$ **M1**

$y = 67$ **A1**

(d) $2\,(2^x) + 3 = 35$

$2^x = 16 = 2^4$ **M1**

$x = 4$ **A1**

FT from (b) in (c) and (d).

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4. [Maximum mark: 5]

(a) $3a + 3b\sqrt{2} + a\sqrt{2} + 2b$ **M1**

$(3a + 2b) + (a + 3b)\sqrt{2}$ **A1**

(b) $3a + 2b = 4$ and $a + 3b = -1$ **M1**

$a = -1 - 3b,\ \ 3(-1 - 3b) + 2b = 4,\ \ -7b = 7$

$a = 2,\ \ b = -1$ **A1A1**

Elimination or substitution both earn M1 here; the method is not prescribed. Common error: $\sqrt{2} \times \sqrt{2}$ *taken as* 4 *gives* $3a + 4b$*; M1A0 in (a).*

5. [Maximum mark: 21]

(a) **A5**

16	32	64	27	81	25	125	$\frac{1}{8}$	$\frac{1}{9}$	$\sqrt{5}$
2^4	2^5	2^6	3^3	3^4	5^2	5^3	2^{-3}	3^{-2}	$5^{\frac{1}{2}}$

A1 for every two correct entries. The base must be 2, 3 *or* 5*:* 4^2 *or* 8^2 *earns no credit for that entry.*

(b) (i) a^6 **A1**

Common error: a^5 *(exponents added) earns A0.*

(ii) $8x^{12}$ **A1**

(iii) $9a^4b^2 \times 2a^{-3}b$ **M1**

$18ab^3$ **A1**

(c) (i) $2^{x+1} = 2^5$ **M1**

$x + 1 = 5$, so $x = 4$ **A1**

(ii) $3^{2x} = 3^{3(x-1)}$ **M1**

$2x = 3x - 3$, so $x = 3$ **A1**

(iii) $5^{2x} = 5^{-3}$ **M1**

$2x = -3$, so $x = -\frac{3}{2}$ **A1**

(d) (i) $3^{2x} = (3^x)^2 = u^2$ **R1**

$u^2 - 10u + 9 = 0$ **A1**

(ii) $(u - 1)(u - 9) = 0$ **M1**

$u = 1$ or $u = 9$ **A1**

(iii) $3^x = 1$, so $x = 0$; $\quad 3^x = 9$, so $x = 2$ **A1A1**

Version (b)

Total 66 marks

1. [Maximum mark: 11]

Elimination is required. Substitution in place of elimination earns M0 on each elimination step; A marks for correct values still apply.

(a) $5x + 2y = 3, \quad 3x + 7y = 25$

$3\times$ first: $15x + 6y = 9; \quad 5\times$ second: $15x + 35y = 125$

Subtracting: $29y = 116$ **M1**

$y = 4$ **A1**

$5x + 2(4) = 3$, so $x = -1$ **A1**

(b) $3x + 2y - z = 1 \;\; (1), \quad 2x - 3y + 4z = 19 \;\; (2), \quad 4x + 5y + 2z = 9 \;\; (3)$

$4 \times (1) + (2)$: $12x + 8y - 4z + 2x - 3y + 4z = 4 + 19$ **M1**

$14x + 5y = 23 \;\; (4)$ **A1**

$2 \times (1) + (3)$: $6x + 4y - 2z + 4x + 5y + 2z = 2 + 9$ **M1**

$10x + 9y = 11 \;\; (5)$ **A1**

$9 \times (4)$: $126x + 45y = 207; \quad 5 \times (5)$: $50x + 45y = 55$

Subtracting: $76x = 152$ **M1**

$x = 2$ **A1**

$14(2) + 5y = 23$, so $y = -1$ **A1**

$z = 3(2) + 2(-1) - 1 = 3$ **A1**

Accept eliminating x or y first; award the same marks for the two two-variable equations, their solution, and the third value. Common errors: scaling only part of an equation (for example $2 \times (1)$ written as $6x + 4y - z = 2$) earns M1A0 for that equation; after eliminating x from (4) and (5), the value found is y, not x.

2. [Maximum mark: 18]

(a) $u_1 + 2d = 14$ **A1**

$\frac{8}{2}(2u_1 + 7d) = 148$, so $2u_1 + 7d = 37$ **A1**

(b) $2\times$ first: $2u_1 + 4d = 28$

Subtracting: $3d = 9$ **M1**

$d = 3$ **A1**

$u_1 = 14 - 2(3) = 8$ **A1**

(c) $u_{20} = 8 + 19(3)$ **M1**

$u_{20} = 65$ **A1**

(d) S_2: $4p - 2q = 2$ **A1**

S_5: $25p - 5q = 35$ **A1**

$2p - q = 1$ and $5p - q = 7$; subtracting: $3p = 6$ **M1**

$p = 2,\ \ q = 3$ **A1A1**

(e) $u_1 = S_1 = 2 - 3 = -1$ **A1**

$u_2 = S_2 - S_1$ **M1**

$u_2 = 2 - (-1) = 3$ **A1**

$d = 3 - (-1) = 4$ **A1**

Common error: $u_2 = S_2 = 2$ *earns M0A0, then FT for* d.

(f) $u_n = -1 + (n-1)(4)$ **M1**

$u_n = -1 + 4n - 4$

$u_n = 4n - 5$ **AG**

Accept $u_n = S_n - S_{n-1} = 2n^2 - 3n - (2(n-1)^2 - 3(n-1))$ *for M1, simplified to* $4n - 5$.

3. [Maximum mark: 9]

(a) $\frac{1}{2}A + B = 4$ **A1**

$4A + B = 11$ **A1**

Subtracting: $\frac{7}{2}A = 7$ **M1**

$A = 2,\ \ B = 3$ **A1A1**

Common error: 2^{-1} *taken as* -2 *gives* $-2A + B = 4$*, so* $A = 1.17$ *(not exact); A0 for that equation.*

(b) $2\,(2^x) + 3 = 2^1 \times 2^x + 3$ **M1**

$= 2^{x+1} + 3$ **AG**

(c) $2^{x+1} + 3 = 131$

$2^{x+1} = 128$ **M1**

$2^{x+1} = 2^7$ **A1**

$x = 6$ **A1**

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4. [Maximum mark: 5]

(a) $2a + 2b\sqrt{3} + a\sqrt{3} + 3b = (2a + 3b) + (a + 2b)\sqrt{3}$ **M1**

$2a + 3b = 5$ and $a + 2b = 4$ **A1**

$a = -2,\ \ b = 3$ **A1A1**

(b) $\frac{5 + 4\sqrt{3}}{2 + \sqrt{3}} = a + b\sqrt{3} = -2 + 3\sqrt{3}$ **A1**

FT from (a). Accept rationalizing with the conjugate $2 - \sqrt{3}$*:* $\frac{(5 + 4\sqrt{3})(2 - \sqrt{3})}{4 - 3} = 10 - 5\sqrt{3} + 8\sqrt{3} - 12$.

5. [Maximum mark: 23]

(a) **A5**

32	64	81	125	$\frac{1}{8}$	$\frac{1}{9}$	$\sqrt{5}$	$4\sqrt{2}$	$\frac{1}{5\sqrt{5}}$	$\sqrt[3]{9}$
2^5	2^6	3^4	5^3	2^{-3}	3^{-2}	$5^{\frac{1}{2}}$	$2^{\frac{5}{2}}$	$5^{-\frac{3}{2}}$	$3^{\frac{2}{3}}$

A1 for every two correct entries.

(b) (i) $8x^{12}$ **A1**

(ii) $9a^4b^2 \times 2a^{-3}b^{-4}$ **M1**

$\dfrac{18a}{b^2}$ **A1**

(iii) $(4a^6)^{\frac{3}{2}} = 4^{\frac{3}{2}}a^9 = 8a^9$ **M1A1**

$\dfrac{8a^9}{8a^3} = a^6$ **A1**

(c) (i) $5^{2(x+1)} = 5^{3(x-1)}$ **M1**

$2x + 2 = 3x - 3$, so $x = 5$ **A1**

Common error: $125 = 5^4$ *earns M0A0.*

(ii) $2^{2x} \times 2^{3(x-1)} = 2^5$ **M1**

$5x - 3 = 5$, so $x = \dfrac{8}{5}$ **A1**

(iii) $3^{3x} = \dfrac{1}{3^2 \times 3^{\frac{1}{2}}} = 3^{-\frac{5}{2}}$ **M1**

$3x = -\dfrac{5}{2}$, so $x = -\dfrac{5}{6}$ **A1**

(d) $9^x = u^2$ and $3^{x+1} = 3 \times 3^x = 3u$ **M1**

$u^2 - 12u + 27 = 0$ **A1**

$(u - 3)(u - 9) = 0$ **M1**

$u = 3$ or $u = 9$ **A1**

$x = 1$ or $x = 2$ **A1A1**

Common error: $3^{x+1} = 3^x + 1$ *or* $u + 1$ *gives* $u^2 - 4u - 4 + 27$*; M0 for the substitution.*